

IMPLEMENTATION OF VARIOUS ARRAY GEOMETRIES IN SMART ANTENNA

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Abstract – Space division multiple access (SDMA) is a promising candidate for improving channel capacity in future wireless communication systems. Since the discrimination performance of the user in the spatial domain depends on the array arrangement, an optimum element arrangement for SDMA should be defined. The pattern of antenna array is determined by array geometry. Two-dimensional spatial filters that can be implemented by microstrip technology are capable of filtering the received signal in the angular domain as well as the frequency domain. This paper focuses on various geometry of eight and nine elements antenna array using circularly patch elements.

I. INTRODUCTION

One of the most recent innovations to overcome the demands for increased capacity nowadays is that of deploying smart antennas for wireless communications [1]. Narrower main beam and more nulls in the pattern can resolve the Signals-Of-Interest (SOI) more accurately and allow the smart antenna system to reject more Signals-Not-Of-Interest (SNOI). Smart antenna system design is subdivided into two major areas: **antenna design and adaptive beamforming algorithm**. The array geometry includes mainly uniform linear array, uniform rectangular array and circular array. Linear array has excellent directivity and it can form **the least main-lobe** in given direction. But it is difficult to keep consistent in large range and do not treat equally all **azimuths** [2]. This paper mainly studies circular array and **multi-ring array**. Circular array is high side-lobe geometry so the distance of array antenna should be small and the basic symmetry of circular arrays offers a great ability to compensate for the effects of **mutual coupling** [3]. **The optimum criterion adopted by this paper is that the geometry must make the main-lobe minimum in the case that the minimum side-lobe is less than half-power.**

This paper analyzes the performance of array pattern based on the eight and nine elements circular array with one main-lobe and studies influence of array structure and various arrays radius impact on the pattern of the antenna array. The elements are circular microstrip patch antenna and they are designed to cover both 1.85 and 2 GHz. These ranges of frequencies are very desirable in modern wireless communications.

II. CIRCULAR PATCH ANTENNA DESIGN FOR 1.85-2 GHz WIRELESS COMMUNICATIONS

In this section, a **circular patch antenna** for wireless communications is characterized in detail. Now we consider design formulas of the circular patch antenna studied in [4]-[5] and list the antenna parameters. The antenna geometry is shown in Figure 1A. The antenna parameters are listed below: (in millimeters)

$$a = 30, h = 1.27, (X_f, Y_f) = (7.3, 0)$$

Where a is radius of the antenna, h is the substrate thickness in mm and the antenna is fed by a coaxial probe. The S_{11} is calculated by IE3D from software and measured on an HP-8510 network analyzer. Some experimental results prove the validity of this design.

III. ARRAY GEOMETRIES

Figure 1B shows the eight elements circular array with radius a as the array elements uniformly distributed in the circumference of circular antenna array. The angular position of the n th element on the array is given by:

$$\Phi_n = 2\pi\left(\frac{n}{N}\right), n = 1, 2, 3, \dots, N \quad (1)$$

Further, assuming that the wave front passes through the origin at the time instant $t = 0$, it impinges on the n th element of the array at the time:

$$\tau_n = -\frac{a}{c} \sin \theta \cos(\Phi - \Phi_n), n = 1, 2, 3, \dots, N \quad (2)$$

Where c is the propagation speed of light in free space. Assumed that θ is the angle between incident direction and the first element of the antenna array. The phase deference φ between the array element and the centre of circle is:

$$\varphi = \frac{2\pi a}{\lambda} \cos\left(\frac{2\pi}{N} n - \theta\right) \quad (3)$$

Where: a is radius of array, N is the sum of array elements and λ is wavelength of signal.

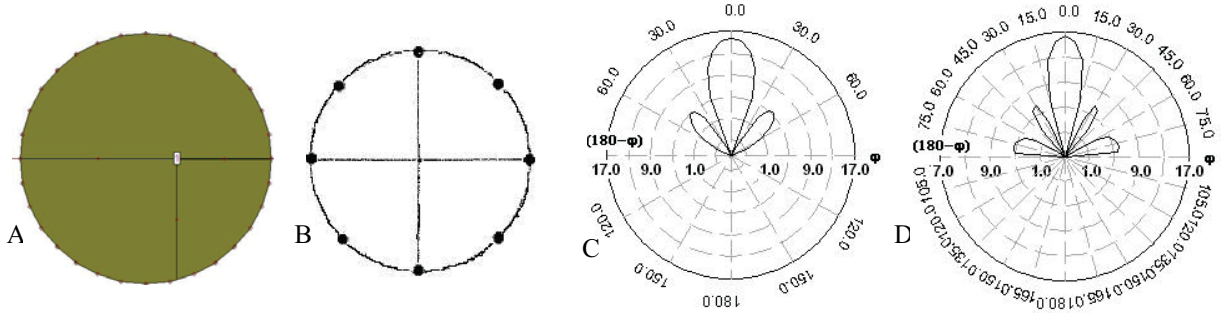


Fig. 1. A: array element, B: shows the eight elements circular array, C, D: array patterns.

A. The eight and nine elements circular array

Figure 1C shows the pattern of the eight elements circular array, defining **MML** as maximum main-lobe width and **MMS** as minimum side-lobe width. When $a=0.5\lambda$, $MML=40^\circ$ and the maximum side-lobe corresponds to the angle $\alpha=45^\circ$, and the power of maximum side-lobe is just less than half-power. Then the performance of the antenna array is optimum. From figure 1D, when $a=0.7\lambda$, $MML=30^\circ$, with increasing of the circle radius, the value of MML decreasing, simultaneously, the number of side-lobe increasing. The signal out of main-lobe cannot be efficiently restrained.

Therefore the width of minimum main-lobe is 30° in the eight elements circular array. But the performance is not ideal. In order to improve the performance characteristic of the antenna array, the multi-layer circular array is used. The pattern function of **concentric circular array** is:

$$F(\theta) = \left| \frac{1}{M \times N} \left(1 + \sum_{i=0}^{N-1} e^{j \frac{2\pi R}{\lambda} i} \left[\cos\left(\frac{\pi}{N} i - \theta_m\right) - \cos\left(\frac{\pi}{N} i - \theta\right) \right] + \dots + \sum_{i=0}^{N-1} e^{j \frac{2\pi R_m}{\lambda} i} \left[\cos\left(\frac{\pi}{N} i - \theta_m\right) - \cos\left(\frac{\pi}{N} i - \theta\right) \right] \right) \right|^2 \quad (4)$$

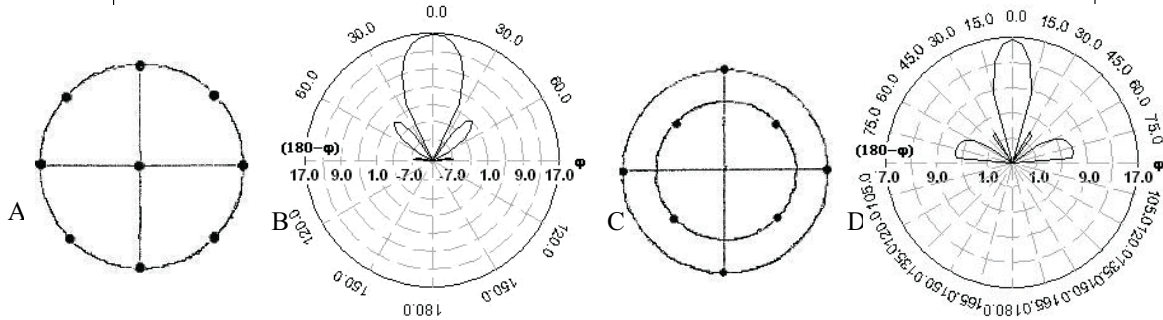


Fig. 2. A: shows the nine elements circular array, C: 4+4 two-ring array geometry, B, D: nine elements array patterns.

Figure 2A shows the nine elements circular array with radius $a=.6\lambda$. In this case $MML=50^\circ$ and the maximum side-lobe corresponds to the angle $\alpha=45^\circ$, but they are so weak. Figures 2B,D show the patterns of the nine elements circular array. When $a=0.8\lambda$, $MW=30^\circ$, the number of side-lobes increasing but they are still weak.

B. 4+4 circular array

Figure 2C indicates the structure of 4+4 two-ring array. This multi-ring array consists of two circular arrays uniformly distributed in the circumference of antenna array. The included angle of the first element of two rings connected with the center of circle is β . During the searching, the outer ring is fixed and the location and direction of the inner ring that relative the outer ring are adjusted. It assumed that the radius of outside circle is

a_1 and the radius of inside circle is a_2 . Figure 3A. indicates the pattern when $a_1 = .9\lambda$, $a_2 = .45\lambda$, $\beta = 45^\circ$ main-lobe is large. Figure 3B shows the best performance when $\beta = 45^\circ$, $a_1 = \lambda$, $a_2 = .7\lambda$

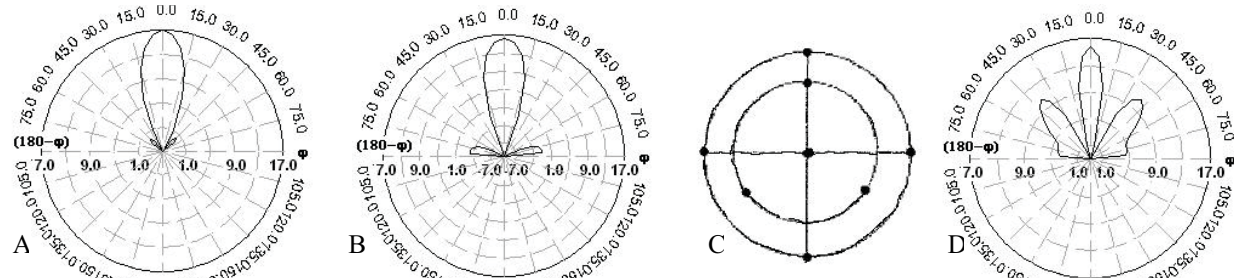


Fig. 3. A,B: 4+4 two-ring array patterns. C: 1+3+4 circular array. D: 1+3+4 circular array pattern.

C. 1+3+4 circular array

The antenna geometry is shown in Figure 3C. In this array a_1 , a_2 and β can be adjusted. Searching under the optimum criterion by increasing a , it shows the best performance when $\beta = 45^\circ$, $a_1 = 1.3\lambda$, $a_2 = \lambda$ and side-lobe is just less than half-power and they are wide. In this case MML = 24° . Best performance is shown in Figure 3D.

D: 3+5 circular array

The outer circular has five antenna elements and the number of array on the inner ring is three. Figure 4A shows the structure. With the radius of the outer ring as a_1 and the radius of the inner ring as a_2 . The outer ring is fixed. Figure 4B indicates the pattern of array when $\beta = 45^\circ$, $a_1 = 1.3\lambda$, $a_2 = \lambda$, and side-lobe is also much larger. Searching under the optimum criterion, it shows the best performance when and side-lobe less than half-power. By comprising some arrays, it is obvious when the elements of outer ring are odd the performance is better than the even case. Figure 4B shows the array pattern.

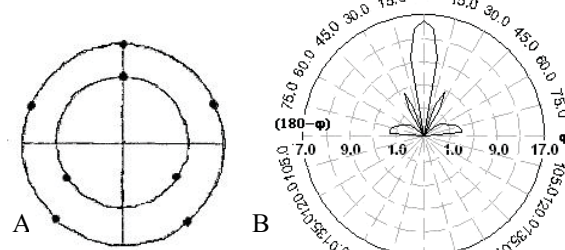


Fig. 4. A: 3+5 two-ring array pattern. B: 3+5 circular array geometry.

IV. CONCLUSION

It is very important for searching the best pattern of antenna array based on the fixed element through changing geometry of antenna array. This paper studies a lot of structure of eight and nine elements antenna array and has the conclusions: 1-The radius of circular array mainly influence main-lobe of pattern Larger radius makes narrower main-lobe of the pattern vice versa. 2-In order to keep performance consistent in total range of 360° , the elements must be kept uniformly around the ring.

REFERENCES

- [1.] M. Chryssomallis, "Smart antennas" *IEEE Antennas Propagat. Mag.*, vol. 42, no. 3, pp 129-136, June 2000.
- [2.] P. Ioannides and C.A. Balanis "Uniform circular arrays for smart antennas" *IEEE Conf*, pp.2796-2800, 2004
- [3.] C.M. Tan, P. Fletcher, M.A. Beach, A.R. Nix, M. Landmann, and R.S. Thoma, "On the Application of Circular Arrays in Direction Finding Part II: Experimental evaluation on SAGE with different circular arrays," in *1st Annual COST 273 Workshop*, Espm, Finland, may 2002, pp. 29-30.
- [4.] C.A. Balanis, *Antenna Theory, Analysis and Design*, 2nd ed., New York, Wiley, 1997.
- [5.] R. Grag, P. Bhartia, I. Bahal, A. Ittipiboon, *Microstrip Antenna Design Handbook*, 2nd ed., Artech House, 2001.